Machine Learning Application in Water Quality Using Satellite Data

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Abstract. Monitoring water quality is a critical aspect of environmental sustainability. Poor water quality has an impact not just on aquatic life but also on the ecosystem. The purpose of this systematic review is to identify peer-reviewed literature on the effectiveness of applying machine learning (ML) methodologies to estimate water quality parameters with satellite data. The data was gathered using the Scopus, Web of Science, and IEEE citation databases. Related articles were extracted, selected, and evaluated using advanced keyword search and the PRISMA approach. The bibliographic information from publications written in journals during the previous two decades were collected. Publications that applied ML to water quality parameter retrieval with a focus on the application of satellite data were identified for further systematic review. A search query of 1796 papers identified 113 eligible studies. Popular ML models application were artificial neural network (ANN), random forest (RF), support vector machines (SVM), regression, cubist, genetic programming (GP) and decision tree (DT). Common water quality parameters extracted were chlorophyll-a (Chl-a), temperature, salinity, colored dissolved organic matter (CDOM), suspended solids and turbidity. According to the systematic analysis, ML can be successfully extended to water quality monitoring, allowing researchers to forecast and learn from natural processes in the environment, as well as assess human impacts on an ecosystem. These efforts will also help with restoration programs to ensure that environmental policy guidelines are followed.

1. Introduction

1.1. Water quality

Water quality describes a state of a water body, as well as its chemical, physical, and biological aspects, including its usefulness for a particular activity (i.e., fishing, swimming or drinking). Substances that can damage aquatic species if found in high enough quantities can also impair water quality. Monitoring water quality is a critical aspect of environmental sustainability. Poor water quality has an impact not just on aquatic life but also on the ecosystem. The following variables are also be used to provide an indicator of water quality: the content of dissolved oxygen (DO); amounts of fecal coliform bacteria from people and animal wastes; levels or ratio of plant nutrients nitrogen and phosphorus; volume of particulate suspended matter (turbidity) and the amount of salt (salinity) in the water. To assess water quality, quantities of substances such as pesticides, herbicides, heavy metals, and other pollutants can be calculated. The abundance of chlorophyll-a (Chl-a), a green pigment

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present in microscopic algae, is often filtered from water samples in many water bodies to provide an indicator of the microalgae living in the water column [1].

1.2. Satellite and remote sensing

Remote sensing is the method of surveying the surface of the earth without making any physical connection. It is used primarily to collect data from the earth's properties and analyze changes in the earth's environment. Along with improvements in satellite technologies and device processing capability, remote sensing has become more widely used in this era. Remote sensing generates spectral, infrared, and radar images that can be interpreted and analyzed to extract useful knowledge about earth elements like water, soil, plants, and the atmosphere, among others. These data are often used to forecast weather and environment, as well as for tracking animal populations, crop health, shoreline changes, and land-use change detection. The resolution of remote sensing data varies depending on the satellite capability. Remote sensing data has recently been produced and effectively utilized to collect water quality information as a solution to the limitations of traditional methods [2]. Remotely sensed data sets are usually more extensive than those collected directly on site by providing better resolution and typically higher temporal frequency and resolution for spatial coverage [3]. Remote satellite sensing examples include Landsat, Sentinel, MODIS, MERIS and VIIRS.

1.3. Machine learning

Machine Learning (ML) is a type of statistical approach that can automatically learn from data and construct a detection, estimation, or classification model that minimizes the variance between the training and prediction datasets without being actively programmed. ML, also known as statistical learning, is providing data to a computer that can be "trained" using known or predetermined attributes or objects to allow semi-automatic or automatic detection, classification, or pattern recognition. ML enabling remotely sensed water quality estimate has grown in popularity in recent years as a result of improvements in algorithm development, computer power, sensor systems, and availability of data [4].

1.4. Systematic review objectives

In this systematic review, the effectiveness of applying ML methodologies were investigated to retrieve water quality parameters from satellite data. Specifically, the objective of studies, the types of satellite data, the ML methodologies, the significance or outcome of the ML application were summarized.

1.5. Nomenclature

Figure 1 provided the list of the abbreviations, acronyms and symbols used in this manuscript.

2. Materials and Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology was used to prepare and report the results of this study [5]. PRISMA is a standard method to give a systematic review of existing research.

2.1. Eligibility criteria

This study focused on peer-reviewed publications that applied ML to estimate water quality parameters with satellite data. The searches for and screen publications focused on three criteria: (1) water quality parameters, (2) ML techniques, and (3) type of satellite.

ADtrees	Alternating Decision Trees
ANFIS	Adaptive Network-Based Fuzzy Inference System
ANN	Artificial Neural Network
AVHRR	Advanced Very High-Resolution Radiometer
BART	Bayesian Additive Regression Tree
BDtrees	Binary Decision Trees
BGA	Blue-Green Algae
BIOC	Bioclim
BR	Band Ratio
BRT	Boosted Regression Tree
C2RCC	Case-2 Regional/Coast Colour
CART	Classification And Regression Trees
CFNN	Cascade Forward Neural Network
Chl-a	Chlorophyll-a
Chl-b	Chlorophyll-b
Chl-c	Chlorophyll-c
CI-OC3M	Global Ocean Algorithm
CNN	Convolutional Neural Network
COD	Chemical Oxygen Demand
COMS	Communication, Ocean and Meteorological Satellite
CZCS	Coastal Zone Color Scanner Experiment
DIN	Dissolved Inorganic Nitrogen
DIP	Dissolved Inorganic Phosphorus
DNN	Deep Neural Network
DO	Dissolved Oxygen
DOC	Dissolved Organic Carbon
OM	D Domain
DSCOVR	Deep Space Climate Observatory
DT	Decision Tree
E. coli	Escherichia Coli
ELM	Extreme Learning Machine
EnMAP	Environmental Mapping and Analysis Program
EPIC	Earth Polychromatic Imaging Camera
ESA	European Space Agency
ET	Extremely Randomized Tree
ETM+	Enhanced Thematic Mapper Plus
FDA	Flexible Discriminant Analysis
fDOM	Fluorescent Dissolved Organic Matter
FFNN	Feed-Forward Neural Network
GA	Genetic Algorithm
GAM	Generalized Additive Models
GAR	Generalized Addictive Regression
GBDT	Gradient Boosting Decision Tree
GBM	Gradient Boosting Machine
GCOM-C	Global Change Observation Mission – Climate
GEE	Google Earth Engine
GLM	Generalized Linear Model
GMM	Gaussian Mixture Model
GNNWR	Geographically Neural Network Weighted Regression
GOCI	Geostationary Ocean Color Imager
GP	Genetic Programming
GPR	Gaussian Process Regression
GWR	Geographically Weighted Regression
HAC	Hierarchical Ascendant Classification
HICO	Hyperspectral Imager for the Coastal Ocean
HMM	Hidden Markov Model
Hyperion	Hyperspectral Imager
IDW	Inverse Distance Weighting
ISS	International Space Station
kNN	k-nearest neighbour
KRR	Kernel Ridge Regression
LightGBM	Light Gradient Boosting Machine
LR	Linear Regression
LSE	Least Square
LSTM	Long Short-Term Memory
MAHA	Mahalanobis Distance
MARS	Multivariate Adaptive Regression Splines
MAXENT	Maximum Entropy
MAXL	Maxlike
MDA	Mixture Discriminant Analysis
MERIS	Medium Resolution Imaging Spectrometer
MERSI	Medium Resolution Spectral Imager
ML	Machine Learning
MLNN	Multilayer Neural Network
MLP	Multilayer Perceptron Neural Network
MLR	Multiple Linear Regression
MNR	Multiple Nonlinear Regression
MODIS	Moderate-Resolution Imaging Spectroradiometer
MPR	Multiple Polynomial Regression
MSI	Multispectral Instrument
OC3M	Ocean Color 3m
OC4Me	Ocean Colour for Meris
OCI	Ocean Color Instrument
OCN	Ocean Color Net
OLCI	Ocean And Land Colour Instrument
OLI	Operational Land Imager
OLR	Ordinary Linear Regression
PCA	Principal Component Analysis
PCR	Principal Component Regression
PLSR	Partial Least Squares Regression
POC	Particulate Organic Carbon
PPC	Photoprotective Carotenoids
PSC	Photosynthetic Carotenoids
RBF	Radial Basis Function
RF	Random Forest
RLR	Ridge Linear Regressor
RPART	Recursive Partitioning and Regression Trees
SC	Specific Conductance
SDD	Secchi Disk Depth
SeaWiFS	Sea-Viewing Wide Field-Of-View Sensor
SEIK	Singular Evolutive Interpolated Kalman
SGLI	Second-Generation Global Imager
SLSTR	Sea And Land Surface Temperature Radiometer
SMOS	Soil Moisture and Ocean Salinity
SNPP	Suomi National Polar-Orbiting Partnership
SOM	Self-Organizing Map
SpecWa	Spectral Remote Sensing Data and Chlorophyll-a Value of Inland Waters.
SSS	Sea Surface Salinity
SST	Sea Surface Temperature
STA	Subsurface Temperature Anomaly
SVM	Support Vector Machine
TM	Thematic Mapper
TN	Total Nitrogen
TP	Total Phosphorus
TSM	Total Suspended Matters
VIIRS	Visible Infrared Imaging Radiometer Suite
XGBoost	Extreme Gradient Boosting

Figure 1. List of the abbreviations, acronyms and symbols.

2.2. Information sources and search

The peer-reviewed publications were searched in three resources: Scopus, Web of Science and IEEE citation databases. The search was restricted to research articles published in English and in peer-reviewed journals or conference proceedings. The following query is constructed with the Boolean operator *AND* and *OR.* The list of queries is shown in Table 1. The searches were run against the title, keywords and abstract of the publications in different databases separately.

Table 1. Keywords used for article search

Keyword set no.	Search Strategy
1	"machine learning" *AND* "satellite" *OR* "ocean colour" *OR* "organic" *OR* "phytoplankton" *OR* "salinity" *OR* "temperature" *OR* "time series" *OR* "water quality" *OR* "suspended" *OR* "CDOM"
2	"ocean colour" *AND* "ocean color" *OR* "forecast" *OR* "forecasting" *OR* "predict" *OR* "prediction"

2.3. Study selection

The eligibility of publications was evaluated and the publications were screened by examining the titles, abstracts and methods, and then obtained eligible publications through reading the full text.

2.4. Data collection and analysis

The data were documented with the objectives, methodologies, environments, problems investigated, language and datasets for each eligible publication. A narrative synthesis of all relevant papers was carried out and arranged based on (1) research goal, (2) ML methodologies, and (3) scientific findings. While the first perspective demonstrated satellite data applications for water quality monitoring, the second view gave an insight into current techniques of study and challenges when applying ML to process and analyze water quality parameters. The third viewpoint showed the lessons that may be drawn from water quality concerns.

2.5. Risk of bias

This systematic review is biased in certain aspects. To begin with, there is a risk of bias in the review process because there is only one reviewer who screens the literature, and the subjectivity of the inclusion and exclusion criteria may influence the selection of relevant articles. Furthermore, throughout the search procedure, the year range was not specified. This implies that the search results are from all years accessible, starting with the earliest publication discovered in the individual databases and ending with the most current (May 2021). Moreover, though the search was limited to three databases, there are many more databases (e.g., Google Scholar, ACM Digital Library) that may contain more material addressing the ML application in water quality utilizing satellite data discussed in this paper.

3. Results

The process of identifying eligible articles is depicted in Figure 2. Initially, the queries returned 1796 publications. After that, the publications were screened to eliminate duplicates. There are 473 duplicates that were removed. The abstracts and titles were read in order to examine the techniques and account for the aforementioned inclusion and exclusion criteria, resulting in the removal of 1196 articles and the retention of 127 for a more in-depth examination. Following the full publication review, 14 studies were excluded due to non-English language publications and studies that were unable to get access to the manuscripts. Finally, 113 publications between the year 2001 until 2021 were included in the systematic review. Table 2 summarizes the publications in terms of their type of satellite used, ML techniques involved, water quality parameters extracted and significance or outcomes of studies.

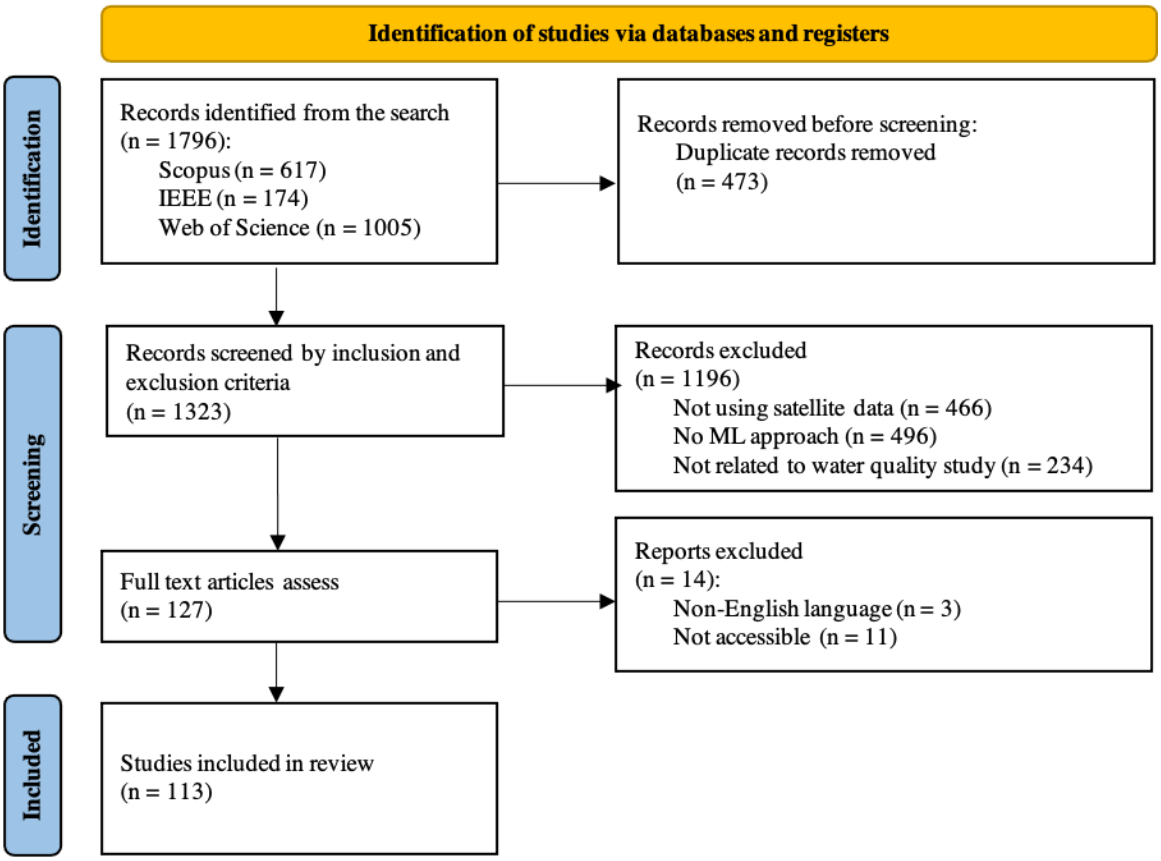


Figure 2. Screening and study identification PRISMA diagram [5].

4. Discussion

The majority of the reviewed studies demonstrated that ML can be effectively applied to learn about water quality monitoring via satellite or remote sensing. This section discusses the insight that can be learned from the reviewed studies.

4.1. Importance of water quality monitoring

A variety of indicators are often used to assess water quality, i.e., turbidity, suspended solids, concentrations of Chl-a, pollution-sediment, DO, CDOM, nutrients (TP, TN, ammonia-nitrogen, nitrate, orthophosphate, silicate), and harmful algae, etc. while water temperature, salinity and many other pollutants are also used as water quality indicators. Nutrient and sediment loads have an impact on water quality. Excess nitrogen and/or phosphorus can lead to eutrophication and fish deaths by increasing algal blooms and aquatic plant growth. The terms suspended-sediment concentration (SSC) and total suspended solids (TSS) are frequently used interchangeably to denote pollution-sediment which is a crucial parameter to consider because of its environmental, economic, and human health implications. [4, 21, 30, 65]. E. coli and cyanobacteria are hazardous organisms that can limit public usage of lakes and coastal waters by lowering dissolved oxygen levels and producing taste and odor problems. Significantly, microcystins, which have been related to liver cancer and tumors in people and animals, have been identified [16, 46, 102, 104]. Monitoring water quality parameters such as Chl-a concentration is crucial in fisheries studies, management, and harvesting since environmental factors impact the number and distribution of fish species for example skipjack tuna [26, 33].

Table 2. Summary of 113 eligible studies.

No.	Satellite/remote sensing data	Water quality parameters involved	Significance/outcomes
1	Sentinel-2	TP, TN, COD	ANN exhibited the best performance followed by RF and SVM regression [6].
2	Sentinel-3/OLCI	Chl-a	SVM regression outperformed ANN, KRR and GPR [7].
3	GOCI	COD, DO, DIP, DIN, oil, pH.	GNNWR outperformed MLP, OLR, GWR and IDW [8].
4	SMOS	salinity	RF produced promising outcomes [9].
5	MODIS/Aqua	temperature	The proposed algorithm which has CNN produced promising outcomes [10].
6	SpecWa	Chl-a	One-dimensional CNN (1D CNN) outperforms RF and BR [11]
7	Sentinel-3/OLCI	Chl-a	LightGBM outperforms the traditional methods [12].
8	SeaWiFS, MERIS, MODIS/Aqua, VIIRS	POC	XGBoost was the most robust among ANN, SVM and BR [13].
9	Sentinel-2, Landsat-8	Salinity	ANN yields the most accurate relationship between salinity concentration and water surface reflectance as compared to ANFIS and MLR models [14].
10	COMS/GOCI, DSCOVR/EPIC, FengYun-3D/ MERSI-II, GCOM-C/SGLI, ISS/HICO, Landsat-8/OLI, MODIS/Aqua, SeaWiFS, Sentinel-2/MSI, Sentinel-3/OLCI, SNPP/VIIRS	Chl-a, CDOM, and suspended solids	MLP was reported significant [15].
11	Sentinel-3/OLCI, Sentinel-3/SLSTR	E. coli, intestinal enterococci, total coliforms, faecal coliforms	Prediction of bathing water quality using DT and kNN on selected bands is more accurate than based on all bands [16].
12	Sentinel-2A/MSI, MODIS	Chl-a	In the winter and summer days, MPR and SVM regression-RBF show the greatest results. Furthermore, LSTM has a lower sensitivity [2].
13	Sentinel-2	Microphytobenthos	RF was reported significant [17].
14	Landsat 8/OLI	Chl-a	XGBoost provides practical approaches [18].
15	SeaWiFS, MODIS/Aqua, VIIRS, OCI	Chl-a	SVM regression model produced promising outcomes [19].
16	SMOS	temperature, salinity	Combining non-linear ANN + HAC produced promising outcomes [20].
17	Sentinel-2A	Chl-a, suspended solids	Cubist bestowed significant accuracy [21].
18	ESA CCI	temperature, salinity, Chl-a	RF outperformed DT and MLP [22].
19	SeaWiFS, MERIS, MODIS/Aqua, VIIRS	Chl-a	The BRT model was used to predict fish species [23].
20	VIIRS	Chl-a	In comparison to in-situ data, RF has a greater accuracy on satellite observations [24].
21	Landsat-8, Sentinel-2	BGA, Chl-a, fDOM, DO, SC, and turbidity	The proposed method, SVM regression, MLR and ELM regression performed significantly on particular parameters [25].
22	AVHRR, AMSR-E, MODIS/Aqua	temperature, Chl-a	The SVM, BRT, RF, MARS, GAM, CART, MLP, RPART, and MAXENT algorithms all performed better than the others, while the FDA, MDA, BIOC, DOM, MAXL, MAHA, and RBF algorithms performed poorly [26].
23	MODIS/Aqua	temperature, salinity, Chl-a	Cubist has been reported as the ideal ML method among ANN, SVM, MLR, MNR and Semi-mechanistic [27].
24	MODIS/Aqua, VIIRS	POC + temperature, salinity, density, spiciness	Both RF regressor and RLR show promising results [28].
25	MERIS, Sentinel-3/OLCI	Chl-a	SVM outperformed the other ML models in other studies [29].
26	MODIS	salinity, Chl-a, suspended solids, temperature	RF and SVM outperformed GLM and GAM [30].
27	MODIS	Chl-a	CNN performs better than SVM regression [31].
28	Sentinel-2A/B	Chl-a	GPR yields better results than SVM, MLP, RFR and MLR [32].
29	MODIS/Aqua	temperature, Chl-a	DT acquires a better performance than GLM [33].
30	FengYun-3D/MERSI-II	Chl-a, suspended solids	MLNN was reported significant [34].
31	Sentinel-2	salinity	ANN gives accurate results for a particular level of atmosphere [35].
32	Landsat-8/OLI	salinity	MLP + GA has had the best performance and accuracy among SVM and OLS [36].
33	GEE, Landsat-8, Sentinel-2	BGA phycocyanin, Chl-a, DO, SC, fDOM, turbidity, and pollution-sediments	DNN and LSTM outperformed PLSR and SVM regression for certain water quality parameters [4].
34	MERIS	Turbidity, SDD	The models performed well with slight outperformance for GPR followed by LR, SVM regression and RF regression [37].

Table 3. Summary of 113 eligible studies (continued).

No.	Satellite/remote sensing data	Water quality parameters involved	Significance/outcomes
35	MERIS	Chl-a	The MLP outperforms the SVM regression to capture satellite Chl-a [38].
36	MODIS/Aqua	salinity, Chl-a, temperature, CDOM	Polynomial regression was reported significant [39].
37	Sentinel-2/MSI	Chl-a	OCN outperformed SVM, GPR, C2RCC, BRT, BR and OC4M [40].
38	GEE	DO, temperature, salinity, Chl-a, and pH	RF bestowed significant accuracy [41].
39	CZCS, SeaWiFS	Chl-a	SVM has high efficiency for the study [42].
40	Landsat-8	Chl-a, TP, TN	The ensemble of ANN, SVM, RF and KNN increased the inversion water quality parameter results [43].
41	Aquarius	salinity	The DNN algorithm outperforms SVM, GPR and KRR in terms of performance [44].
42	MODIS	Chl-a	SVM regression showed the best performance among ANN, RF, CI-OC3M, MLR, GAR, PCR and GBM [45].
43	Sentinel-3/OLCI	cyanobacterial	Hierarchical Bayesian Spatio-temporal modeling shows high performance with low Deviance Information Criterion (DIC) value [46].
44	VIIRS	Chl-a	SVM method produced promising outcomes [47].
45	Sentinel-3/OLCI	Chl-a	GPR demonstrated to have strong generalization capabilities [48].
46	MODIS/Aqua	temperature, salinity, Chl-a	The RF regression ensemble-based model showed satisfactory performance among SVM, DT, MLP, MLR, PCR and MNR [49].
47	MODIS, VIIRS	temperature	The ADtrees classifier outperformed the BDtrees [50].
48	Landsat-5, Landsat-8, Sentinel-2, Sentinel-3, EnMAP and Hyperion	Chl-a	The performance of ANN, SVM, RF and MARS in terms of regression is comparable [51].
49	MODIS/Aqua	Chl-a	XGBoost produced promising outcomes [52].
50	Sentinel-1, Sentinel-2	Suspended solids, dissolved solids	SVM outperforms RF, DT and kNN [53].
51	Landsat-5, Landsat-7, Landsat-8	suspended solids, Chl-a, turbidity	ANN outperforms SVM, RF and Cubist [54].
52	AMSR-2, SMOS	salinity	RF model outperforms GBDT model [55].
53	MODIS	Chl-a	The RF regression ensemble produced promising outcomes [56].
54	Landsat-8, GEE, Sentinel-2	turbidity, suspending solids, TP, TN	SVM provided higher accuracy than ANN [57].
55	MODIS/Terra	Turbidity, temperature	ANN relatively good accuracy [58].
56	SeaWiFS, MERIS, MODIS/Aqua	Chl-a, temperature	ET model shows better performance than the RF model [59].
57	MODIS/Terra, Landsat-7/ETM+	Chl-a, cyanobacterial	RF model produced promising outcomes [60].
58	MODIS/Aqua	Chl-a	LR was used to compare retrieval algorithm, OC3M [61].
59	VIIRS, MODIS, SeaWiFS	temperature, salinity, Chl-a	ANN demonstrates a very good ability to generalize in terms of both space and time [62].
60	AVHRR	temperature, salinity	ANN technique successfully predicts early drift paths of green tides [63].
61	Landsat 8/OLI	Chl-a, suspending solids, TP, TN	Multiple Regression was reported significant [64].
62	MODIS	Suspended solids	The best result was achieved by RF, which was followed by SVM, GAM, GLM, BART, MARS and CART [65].
63	SeaWiFS, AVHRR	Chl-a, temperature	The best algorithm was RF, which was followed by SVM, ANN, and PLSR [66].
64	Landsat-5, Landsat-7	Chl-a, CDOM, suspended solids	The BRT model predictive performance was significantly better than MLR [67].
65	Sentinel-2	algal	LSE, RBF and ANFIS have good performance proving the reliability and accuracy [68].
66	MODIS/Aqua, MERIS	Chl-a, CDOM	GPR and SVM regression models are confirmed to show stronger regression power than PLSR [69].
67	MODIS/Aqua	Chl-a, temperature	Multivariate Regression model was reported significant [70].
68	Landsat-8/OLI	Chl-a	SVM regression showed slightly superior performance than ANN [71].
69	SeaWiFS, MODIS/Aqua, MODIS/Terra	CDOM, POC, DOC	MLR outperform RF tree-bagger [72].
70	Sentinel-3/OLCI	Chl-a, CDOM, suspended solids	GPR was reported significant [73].
71	Sentinel-3/OLCI	Chl-a, CDOM, suspended solids	SVM, RF, KRR and GPR are very efficient except for RLR [74].
72	SMOS	Temperature, salinity	RF outperformed SVM regression [75].
73	Landsat-4, Landsat-5, Landsat-7/ETM+, Landsat-8/OLI	Suspended solids	By a significant margin, ELM outperformed FFNN and CFNN, as well as RF and SVM [76].
74	MODIS/Aqua	temperature	ANN produced promising outcomes [77].
75	SeaWiFS, MODIS/Aqua, MERIS, VIIRS	suspended solids	The validation of the generated product is quite important when using HMM [78].
76	MERIS, MODIS/Aqua, SeaWiFS	Chl-a	GMM uses for clustering and assimilated into the 1D models using the SEIK filter [79].
77	MERIS, MODIS/Aqua, SeaWiFS	salinity	MLP showed the best performance among SVM, RF and DT [80].

Table 4. Summary of 113 eligible studies (continued).

No.	Satellite/remote sensing data	Water quality parameters involved	Significance/outcomes
78	GOCI	temperature, salinity, Chl-a, CDOM	RF generally performed better than SVM regression [81].
79	VIIRS/NPP, MODIS/Terra, MODIS/Aqua	Chl-a, TP, TN, SDD	ANN strongly demonstrates effectiveness and reliability [82].
80	AVHRR, SeaWiFS	Chl-a, temperature	BRT showed slightly higher prediction performance than GAM and GLM [83].
81	VIIRS, MERIS, MODIS/Aqua, SeaWiFS	Chl-a	BRT performs the best in terms of prediction [84].
82	MODIS/Aqua, VIIRS	Chl-a	MLP produced promising outcomes [85].
83	MODIS	Chl-a	SVM outperforms the classical approach [86].
84	GOCI	Phytoplankton, suspended solids, CDOM	SVM produced promising outcomes [87].
85	MERIS, MODIS	Chl-a	SVM combine with Linear, polynomial, RBF, sigmoid regression analysis improves the precision of the algorithm [88].
86	MODIS/Aqua	Salinity	GAM outperformed an ANN [89].
87	VIIRS	temperature, salinity, Chl-a	MLP method produced promising outcomes [90].
88	Landsat-5/TM	suspended solids	LR and quadratic models perform better than Logarithmic, Power and Exponential [91].
89	MERIS, MODIS	Chl-a	LR, polynomial, exponential functions and PCA were used for the partitioning mechanism. SVM method is used for the iterative classification process [92].
90	SMOS, Aquarius	salinity, temperature	SVM produced promising outcomes [93].
91	MODIS	suspended solids	MLP method produced a superior performance [94].
92	MODIS/Terra	total organic carbon (TOC)	The ANN model was chosen among GP and ELM for the forecasting method [95].
93	Landsat-5/TM, Landsat-7/ETM+, MODIS/Terra	Chl-a, SDD	GP shows satisfactory results [96].
94	AVHRR, SeaWiFS, MODIS/Aqua	Chl-a, CDOM, temperature,	ANN was used for classification data and MLR was used to reconstructed pCO_2 [97].
95	MERIS	Chl-a, Chl-b, Chl-c, PSC, PPC	MLR significant on ship-based reflectance measurements [98].
96	MODIS	Chl-a	Algorithms using SVM are able to give better results than DT and Log-linear [99].
97	MODIS/Terra, MERIS, Landsat/TM	microcystin	GP use for data mining purpose [100].
98	GOCI	Chl-a, suspended solids	SVM regression outperformed RF, Cubist [101].
99	Landsat, MODIS	microcystin	GP model performed slightly better than ANN [102].
100	MODIS	TP	GP use for spatiotemporal mapping [103].
101	MODIS/Terra, Landsat-5, Landsat-7	microcystin	GP was able to produce a positive correlation with the observed results [104].
102	MODIS/Aqua	Salinity, Chl-a, temperature	ANN was reported significant [105].
103	MODIS/Terra	SDD, suspended solids, Chl-a	GP advantage was identified and been selected among ANN and MLR for estimation [106].
104	MODIS	Chl-a, temperature	Bayesian network produced promising outcomes [107].
105	SPOT-5	COD, ammonia-nitrogen, permanganate index (COD_{Mn})	Combine the GA–SVM regression model is significantly better than MLR [108].
106	MERIS	Chl-a	MLP was able to show prediction performance [109].
107	MODIS, SeaWiFS, AVHRR, GEE	Chl-a, CDOM, temperature, salinity, suspended solids	Gaussian use for creating the 3D surface optical forecast [110].
108	MERIS	Suspended solids, Chl-a	SVM use for estimation [111].
109	SeaWiFS	Suspended solids, Chl-a, nitrate, orthophosphate, silicate, salinity, temperature	In terms of determining temporal and spatial variability, MLR performed well [112].
110	SeaWiFS, AVHRR	Chl-a, temperature	SOM has produced robust estimation [113].
111	SeaWiFS	CDOM, suspended solids, temperature, salinity	MLR produced promising outcomes [114].
112	SeaWiFS	Chl-a, temperature	"Broken-stick" regression able to show prediction performance [115].
113	AVHRR	suspended solids, SDD	LR was reported significant [116].

4.2. Remote sensing for water quality

Optical and thermal sensors collect water quality information with a great spectrum and spatial resolution. Watershed scale models based on ocean color satellite data have been constructed for determining optical active components (OAC) such as Chl-a, suspended solids and CDOM. However, existing satellites cannot directly monitor all water quality parameters, including nutrient concentrations, DO and COD levels, and microorganisms/pathogens, because some of these variables are not optically active, or because there is an absence of hyperspectral data at precise spatial resolutions. Therefore, some studies used OAC as a proxy to estimate non- OAC parameters by determining their relationship [57] and also use possible band compositions from satellite imagery bands [6].

4.3. Machine learning application

Numerous studies have been conducted to determine water quality using satellite data. The majority of the research relied on empirical relationships between satellite-derived reflectance and target water quality parameters to apply relatively simple linear or nonlinear regressions on satellite data. The empirical models produced have a limitation in that they may not operate effectively in diverse environments (such as the open ocean, coastal, river or inland waters). As a result, additional in-situ data is required, as well as parameter values that have been optimized [101]. Moreover, numerous ML models, which are sophisticated nonlinear data-driven approaches that have been tested and widely utilized. Some studies applied ML model comparison and select the best performance ML method to implement for their research. Other studies use the ML method to make improved measuring techniques such as fluorescence line height measurement [7], SSS measurement [9], estimation of POC [13], reduce spectral noise [19], reconstruct missing value [24] and atmospheric correction [35].

4.4. Limitation

This systematic review has numerous limitations that should be acknowledged. Firstly, ML-related keywords included in the search queries are not enough to cover as many related publications as possible. Therefore, this process might miss some studies that failed to be retrieved. Secondly, there is no review of performance used in method evaluation for ML. Thirdly, this review does not includes bibliometric analysis to show the research trends.

5. Conclusion

This systematic review summarized how ML has been applied on satellite data to study water quality issues. The initial search process resulted in 1796 publications, and by refining the search by removing 473 duplicates publication, excluded 1196 non-related topics publications. Through the screening of 127 publications, 113 papers have been selected for data extraction and synthesis. Results also showed that there is a huge variety of ML methods suggested especially on the retrieval of water quality parameters. The most common ML approaches were ANN, SVM, RF, DT, MLP, cubist and GP for monitoring water quality at regional and global scales. According to the systematic analysis, ML can be successfully extended to water quality monitoring, allowing researchers to forecast and learn from natural processes in the environment, as well as assess human impacts on an ecosystem. These initiatives will also aid policymakers and water resource managers in taking proactive actions to prevent the negative consequences of water pollution through restoration projects, as well as ensure that environmental regulatory rules are followed.

Disclosure statement

The authors declared that there is no conflict of interest that could have influenced any part or the entirety of the work reported in this manuscript.

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